

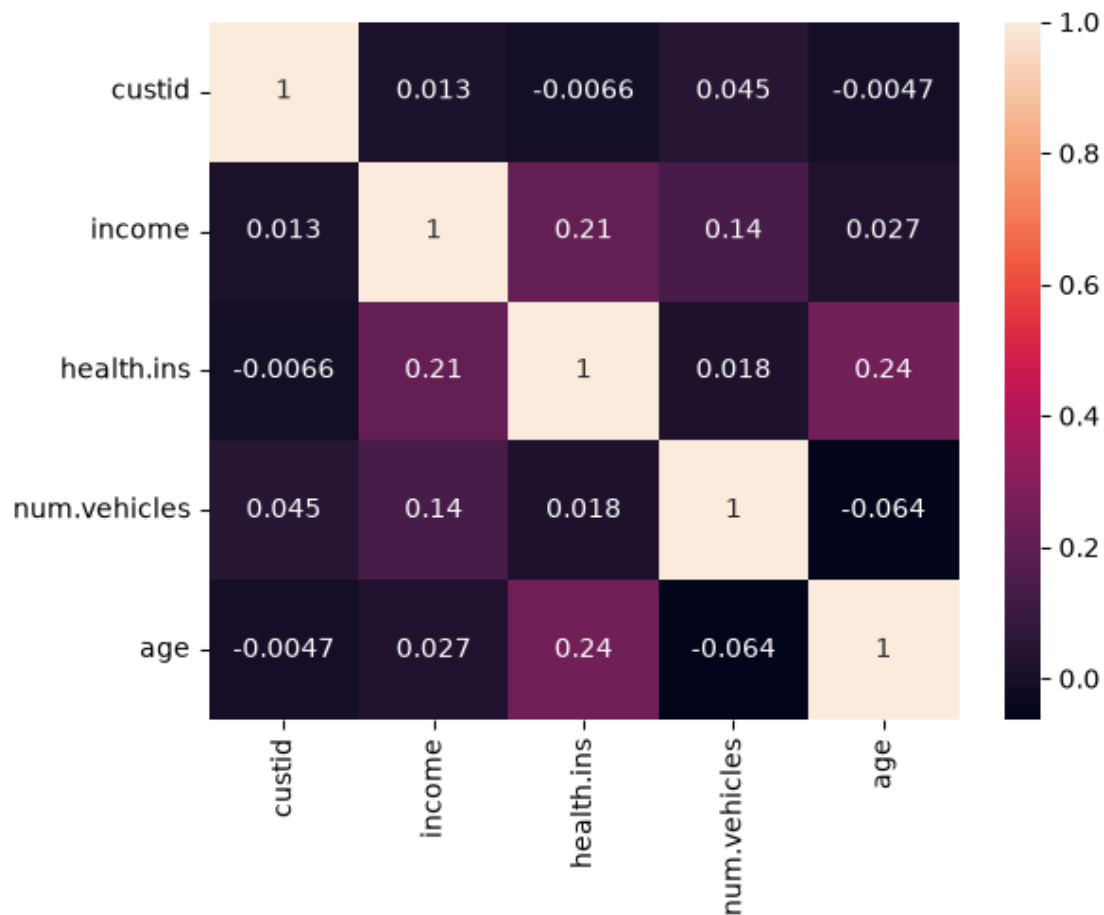
## Module5\_Lab3

October 3, 2026

```
[1]: import pandas as pd
import seaborn as sns
df = pd.read_csv('custdata.tsv', sep='\t')

sns.heatmap(df.corr(numeric_only=True), annot=True)
```

[1]: <Axes: >

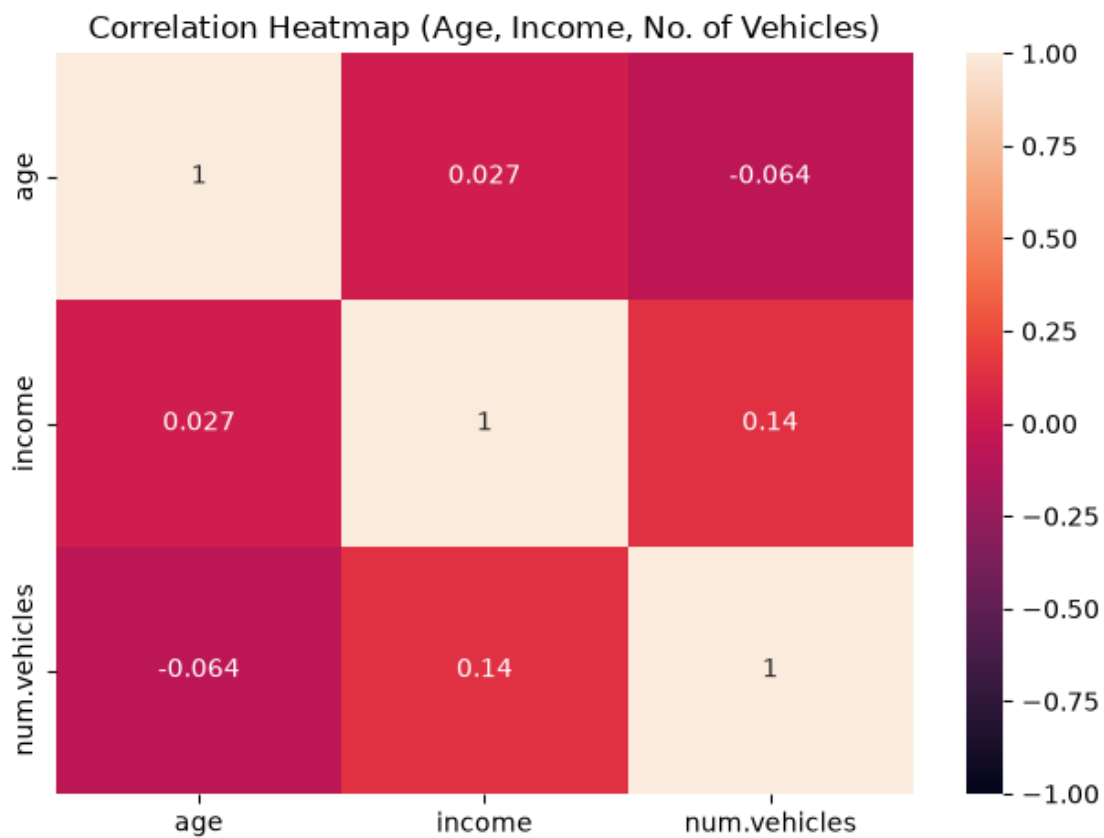


```
[2]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('custdata.tsv', sep='\t')

cols = ['age', 'income', 'num.vehicles']
corr_matrix = df[cols].corr(method='pearson')

# Plot correlation heatmap
plt.figure
sns.heatmap(corr_matrix, annot=True, vmin=-1, vmax=1)
plt.title('Correlation Heatmap (Age, Income, No. of Vehicles)')
plt.tight_layout()
plt.show()
```



```
[3]: r = df['age'].corr(df['income'])
print(f"Pearson's r: {r:.4f}")
```

Pearson's r: 0.0274

No correlation between age and income

```
[4]: r = df['income'].corr(df['num.vehicles'])  
     print(f"Pearson's r: {r:.4f}")
```

Pearson's r: 0.1392

There is a relationship between income and number of vehicles but it is weak.

```
[5]: r = df['age'].corr(df['num.vehicles'])  
     print(f"Pearson's r: {r:.4f}")
```

Pearson's r: -0.0637

Correlation is negative.

Among the three, only income and number of vehicles appear to have a correlation.